# Math SL Internal Assessment
# Genshin Impact Damage Optimisation

## 1. Introduction

Genshin Impact is a third-person action game where players build teams of characters, progress through the game's story, and clear combat challenges. Over my three years of playing, I have spent a lot of time trying to make my characters deal as much damage as possible to clear the hardest game modes: Spiral Abyss, Imaginarium Theater, and Stygian Onslaught. This process, typically called build optimisation, involves choosing the right equipment for each character.

My main damage dealer is Neuvillette. His damage depends on several mechanics at once: a passive ability that changes his damage bonus as his HP(health points) drops during an attack, an artifact set that changes his critical hit rate as his HP fluctuates, and a damage formula that scales directly with his maximum HP. Together, these create a complex system in which the mechanics interact, and all affect the final damage.

**Aim**: The aim of this exploration has two parts: to model the probability distribution of artifact substat rolls and determine what CRIT stats are realistically achievable, and to use a mathematical damage model to find the optimal CRIT Rate to CRIT DMG ratio, and optimal goblet main stat for Neuvillette. The purpose of this investigation is to mathematically optimise Neuvillette's damage using calculus, probability, and statistics in order to clear more combat challenges

## 2. Game Mechanics Overview

Every hit in Genshin Impact follows the same general formula (Genshin Impact Wiki, Damage):

$$DMG = Base\ DMG \times (1 + DMG\ Bonus) \times DEF\ Multiplier \times RES\ Multiplier \times CRIT\ Multiplier$$

Base DMG is the raw damage before any modifiers. For Neuvillette, his main attack scales with his maximum HP, so:

$$Base\ DMG = M \times maxHP \quad \text{(M = 12.51\% per tick at talent level 8)}$$

where M is the talent multiplier, and maxHP is his total maximum HP.

DMG Bonus is the sum of all percentage damage bonuses from equipment and passive abilities, expressed as a decimal (for example, +30% DMG Bonus adds 0.30). These bonuses are added before being multiplied by other terms in the formula.

DEF Multiplier accounts for the enemy's defence. The formula is (Genshin Impact Wiki, Damage):

$$\frac{Lc+100}{(Le+100)+(Lc+100)} = DEF\ Mult$$

where Lc is the character's level, and Le is the enemy's level. For Neuvillette at level 80 fighting a level 100 enemy (standard for endgame content like Spiral Abyss):

$$\frac{80+100}{(100+100)+(80+100)} = 0.474$$

RES Multiplier accounts for the enemy's elemental resistance. For a typical enemy with 10% Hydro RES:

$$RES\ Mult = 1 - 0.10 = 0.90$$

The CRIT Multiplier is where CRIT Rate (CR) and CRIT DMG (CD) factor in. CRIT Rate is the probability that a single hit deals a critical strike. CRIT DMG is the percentage bonus to damage when a critical strike occurs. A non-critical hit has a multiplier of 1, and a critical hit has a multiplier of (1 + CD). Since each hit is an independent event with the same probability of success, CR, the expected value of the CRIT Multiplier is:

$$CRIT\ Mult = (1 - CR) \times 1 + CR \times (1 + CD) = 1 + CR \times CD$$

Putting it all together, the expected damage per tick is:

$$D(CR,\ CD) = K \times (1 + CR \times CD)$$

In this equation, K consists of every other term that is not the CRIT multiplier: the talent scaling, the max HP, all additive damage bonuses, and both defensive and resistance multipliers. Since these variables are not easily changeable, K can be treated as a constant.

The goblet comparison (Section 4.5) changes which goblet main stat is used, which changes K since different goblets change either max HP or DMG Bonus, but the CRIT optimisation that uses substats is unchanged regardless of which goblet is equipped, since the optimal CR and CD ratio depends only on the substat budget, not on main stats. The two questions can therefore be answered independently and combined at the end.

Neuvillette's passive talent, "Discipline of the Supreme Arbitration", adds a Hydro DMG Bonus that depends on his current HP percentage. The exact in-game text reads:

"For each 1% of Neuvillette's current HP greater than 30% of Max HP, he will gain 0.6% Hydro DMG Bonus. A maximum bonus of 30% can be obtained this way."

Expressing this as a function of h, where h is the current percent of HP:

$$PassiveBonus(h) = 0.3 \qquad \text{for } h \geq 0.80$$

$$PassiveBonus(h) = 0.006 \times (100h - 30) \qquad \text{for } 0.30 < h < 0.80$$

$$PassiveBonus(h) = 0 \qquad \text{for } h \leq 0.30$$

The cap of 30% is reached at h = 0.80 (exactly 80% HP):

$$0.006 \times (80 - 30) = 0.30$$

During a full Charged Attack, HP drains from 100% down to approximately 60%. This creates two different phases for the attack:

Phase 1 (100% → 80% HP): Bonus is at its maximum of 30% throughout.

Phase 2 (80% → ~60% HP): Bonus falls continuously from 30% to ~18%:

Artifacts are the main equipment system in Genshin Impact. There are five slots: Flower (always gives flat HP as its main stat), Feather (always gives flat ATK), Sands, Goblet, and Circlet. Each piece also has four substats chosen from a shared pool.

The Sands, Goblet, and Circlet have randomly assigned main stats and substats drawn from a weighted pool specific to each slot. Substats are drawn one at a time from a pool of ten possible stats, excluding whichever stat is already the main stat. Each substat has a weight that controls the probability of being chosen (Genshin Impact Wiki, Artifact/Distribution):

| Substat | Flat Hp | Flat ATK | Flat DEF | HP% | ATK% | DEF% | Energy Recharge% | Elemental Mastery | Crit Rate% | Crit DMG% |
|---|---|---|---|---|---|---|---|---|---|---|
| weight | 6 | 6 | 6 | 4 | 4 | 4 | 4 | 4 | 3 | 3 |

CRIT Rate% and CRIT DMG% each have the lowest weight in the pool. On a Flower, for example, the total pool weight is 3+3+4+4+4+4+4+6+6 = 38 (flat HP excluded as main stat). The chance of any one draw being a specific crit stat is only 3/38 ≈ 7.89%. Because only 4 substats are chosen, most artifact pieces will not have both CRIT Rate and CRIT DMG as substats.

The four substat draws happen without replacement. Once a substat is chosen, it is removed from the pool for that piece. Each time a substat is rolled (either at the start or during an upgrade), it lands on one of four value tiers with equal probability (25% each). For CRIT Rate% and CRIT DMG%, the tiers are (Genshin Impact Wiki, Artifact/Stats):

| Tier | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Crit Rate% | 3.89% | 3.50% | 3.11% | 2.72% |
| Crit DMG | 7.77% | 7.00% | 6.22% | 5.44% |

Every CRIT DMG% tier value is exactly twice the corresponding CRIT Rate% tier value. The average roll for each stat is:

$$avrCR = (3.89 + 3.50 + 3.11 + 2.72) / 4 = 13.22 / 4 = 3.305\%$$

$$avrCD = (7.77 + 7.00 + 6.22 + 5.44) / 4 = 26.43 / 4 = 6.608\%$$

So $avrCD = 2 \times avrCR$ exactly. This ratio is how the CRIT Value metric is defined.

An artifact starts with either 3 substats (80% of pieces) or 4 substats (20%) and are upgraded 5 times total. At each event, one existing substat is selected at random (each substat has a 25% chance) and receives one additional roll at a random tier. For pieces that start with 3 substats, the first event instead adds the fourth substat from the remaining weighted pool.

Because each upgrade picks uniformly from 4 substats, the number of times any given substat gets upgraded follows a Binomial distribution: BinomPDF(5, 0.25, X), where X is the number of rolls into a single stat. This makes hitting a high CRIT value on any single piece unlikely, which is exactly what the simulation in Section 3 shows.

Marechaussee Hunter is the recommended artifact set for Neuvillette. Its bonuses are (Genshin Impact, in-game set description):

2-piece: Normal and Charged Attack DMG +15%

4-piece: "When current HP increases or decreases, CRIT Rate will be increased by 12% for 5s. Max 3 stacks."

The 4-piece bonus interacts with Neuvillette's HP drain mechanic. Every 0.5 seconds during his Charged Attack reduces his current HP by 8%, which counts as an HP change event. Combined with the fact that prior to the attack, Neuvillette heals himself 3 times, this means all three stacks are created and kept throughout the attack, adding +36% CRIT Rate to his effective total.

## 3. ARTIFACT RNG MODELLING

### 3.1 Probability of Double-Crit Substats:

Each substat is taken from the weighted pool one at a time, without replacement. The probability of drawing a specific stat on a given draw equals its weight divided by the total remaining weight at that moment.

**STEP 1 Probability of one permutation**

To model this, let's take a Flower as an example. Flat HP is always the main stat and is excluded from the pool, leaving nine substats with total weight W = 38. Both CR% and CD% have a weight of 3. The probability that CR% is drawn first and CD% is drawn second is

$$P(CR\%\ 1st \cap CD\%\ 2nd) = 3/38 \times 3/35 = 9/1330 \approx 0.68\%$$

This is because after CR% is drawn, its weight of 3 is removed from the pool. The remaining pool weight is 38 − 3 = 35.

This covers only one specific ordering of the four draws. In reality, CR% and CD% can land in any two of the four substat slots, with any two other stats filling the remaining slots. The two filler substats are drawn from the seven non-crit stats: flat ATK and flat DEF (weight 6 each) and HP%, ATK%, DEF%, ER%, EM (weight 4 each).

**Step 2: Adding all valid orderings**

A piece has 4 substat slots filled one at a time. Once we decide which 4 stats should appear, those stats can be in any order across the 4 substats. The number of ways to arrange 4 distinct items in a sequence is 4 × 3 × 2 × 1 = 4! = 24 permutations.

For every set of four substats that includes both CR% and CD%, each of those 24 orderings must be evaluated using the same method as Step 1, and their probabilities are added. When the crit stats appear in later positions, a filler stat must have been drawn first, so the denominator, which is the current weighted pool when crit is selected, is smaller than if crit were selected first. This is why adding up the probabilities of all acceptable permutations is necessary.

Next, we need to know which sets of 4 substats can contain both CR% and CD%. The other 2 slots must be filled by 2 of the 7 non-crit stats. The number of combinations to choose 2 stats from 7 (where only the pair matters, not which comes first) is:

$$C(7,2) \;=\; 7!\;/\;(2!\times 5!) \;=\; (7\times 6)\;/\;2 \;=\; 21 \text{ possible filler pairs}$$

The 7 non-crit stats fall into two weight tiers:

Weight 6: flat ATK, flat DEF (2 stats)

Weight 4: HP%, ATK%, DEF%, ER%, EM (5 stats)

A filler pair can be drawn from these tiers in three distinct ways:

Case 1: 6, and 6: both fillers from the weight-6 tier

$$C(2,2) \;=\; 1\,pair\text{: \{flat ATK, flat DEF\}}$$

Case 2: 6, and 4: one filler from each tier

$$2\times 5 \;=\; 10\,pairs\text{: \{flat ATK, HP\%\}, \{flat DEF, ER\%\}, …}$$

Case 3: 4, and 4: both fillers from the weight-4 tier

$$C(5,2) \;=\; (5\times 4)\;/\;2 \;=\; 10\,pairs\text{: \{HP\%, ATK\%\}, \{ER\%, EM\}, …}$$

$$Total\,Pairs \;=\; 1 \;+\; 10 \;+\; 10 \;=\; 21\,pairs$$

The reason why we can group the substats by weight class is because the probability of a given permutation depends only on the weights drawn at each step, not on which specific stat was chosen. For example, the ordering where CR% comes 2nd and HP% comes 1st gives the same probability as CR%

2nd and DEF% 1st, since HP% and DEF% share the same weight of 4. Every pair within a class, therefore, produces the same total probability when you sum its 24 permutations.

For the (6,6) case, which has 1 pair {flat ATK, flat DEF}, here are two of the 24 orderings, to show why the denominators differ:

CR% 1st, CD% 2nd, flat ATK 3rd, flat DEF 4th:

$$P = 3/38 \times 3/35 \times 6/32 \times 6/26 = 0.029\%$$

flat ATK 1st, CR% 2nd, CD% 3rd, flat DEF 4th:

$$P = 6/38 \times 3/32 \times 3/29 \times 6/26 = 0.035\%$$

Drawing a weight-6 stat first removes 6 from the pool and changes all subsequent denominators, and summing all 24 orderings for this one pair gives 0.8718%. With only 1 pair in the case:

$$Case\ (6,6)\ probability = 1 \times 0.8718\% = 0.8718\%$$

For case (6,4), 10 pairs: Summing all 24 orderings for one representative pair {flat ATK, HP%} gives 0.5188%, and with 10 pairs, we see:

$$Case\ (6,4)\ probability = 10 \times 0.5188\% = 5.1881\%$$

For case (4,4), 10 pairs: One representative pair {HP%, ATK%} has a probability of 0.3096%. With 10 pairs, we see:

$$Case\ (4,4)\ probability = 10 \times 0.3096\% = 3.0964\%$$

To find the total double crit probability for a Flower, we add these probabilities together:

$$P = 0.8718\% + 5.1881\% + 3.0964\% = 9.1562\%$$

Because evaluating all orderings by hand is prone to errors, these sums for the different artifacts were verified using a Python script.

After finding the probability for double crit for all artifact types, the data can be seen in the following table:

| Artifact, and main stat | Flower with flat HP | Feather with flat ATK | Sands with HP% | Goblet with Hydro DMG | Circlet with Crit DMG% |
|---|---|---|---|---|---|
| P(double-crit) | 9.16% | 9.16% | 8.34% | 6.69% | 0% |

For Neuvillette's Circlet, CD% is the main stat and is excluded from the substat pool entirely. Only CR% remains as a possible crit substat, so P(double-crit and CD% main) = 0% regardless of how many pieces are tested.

**3.2 Artifact roll distribution**

Once the four substats are set, the piece gains five upgrade events at +4, +8, +12, +16, and +20. At each event, one of the four substats is selected uniformly at random (25% each) and receives one additional roll.

The number of upgrades a specific substat receives over five events follows a Binomial distribution, which can be modeled by:

$$BinPDF(5,\ 0.25,\ X)\ for\ 0 \leq X \leq 5$$, where X is the number of upgrades

Therefore, if an artifact has 2 crit substats, on any given roll, there is a 50% chance that a crit stat is upgraded. The probability of getting 3 or more upgrades into crit Substats, which is generally acceptable for an artifact, happens 50% of the time, as shown by adding up the chances of rolling into crit 3, 4 and 5 times

$$BinomPDF(5,\ 0.5,\ 3)\ +\ BinomPDF(5,\ 0.5,\ 4)\ +\ BinomPDF(5,\ 0.5,\ 5)\ =\ 0.5$$

Combined with the ~9.2% chance of double-crit in (for a Flower), this means an artifact with at least 3 upgrades in crit substats occurs in about 9.2% × 50% ≈ 4.6% of pieces, showing that getting a piece with usable crit substats can be extremely rare.

**3.3 25 million artifact Simulation and CV Distribution**

Based on all the game mechanics previously mentioned, I made a program to simulate any number of artifacts and to record their stats. In the simulation, CR% and CD% substat values are recorded per piece in order to confirm the mathematical model of artifact rolling.

Crit Value (CV) metric:

To combine CR% and CD% into a single score, we can use:

$$CV\ =\ k \times CR\%\ +\ CD\%$$

In this relationship, k weights CR% relative to CD%. From Section 2, the average substat roll values are avrCR = 3.305% and avrCD = 6.608%, giving:

$$k = avrCD / avrCR = 6.608 / 3.305 = 2.0$$

(The slight discrepancy is due to displaying rounded numbers in the game's code)

So CV = 2 × CR% + CD%. This metric shows that one CR roll is exactly half of one CD roll, and the constraint CV = 2 × CR + CD represents the roll investment. Any two builds with the same CV represent the same total of crit substat value.

Key findings (25,000,000 artifact simulation per piece)

P(CV = 0)(no crit substats):

| | |
|---|---|
| Flower and feather: | ~40.8% |
| Sands: | ~42.7% |
| Goblet: | ~44.7% |
| Circlet (CD% main): | ~68.1% |

On average, over the 5 pieces, only about 1 in 2 pieces have any crit substats. For a CD% main Circlet, 2 in 3 pieces have no crit substats at all, since the only available crit substat is CR%.

Mean CV:

| | |
|---|---|
| Flower and feather: | ~9.26 |
| Sands: | ~8.89 |
| Goblet: | ~8.53 |
| Circlet (CD% main): | ~7.89 |

These low means are largely driven by the ~40–68% of pieces with CV = 0. Among pieces with at least one crit substat, mean CV rises to ~15.2–15.6 for Flower, Feather, Sands, and Goblet.

Along with this, the probability for a piece to have 2 crit substats was recorded.

| | |
|---|---|
| Flower: | ~ 9.15% |
| Feather: | ~ 9.18% |
| Sands: | ~8.35% |

Goblet: ~6.64%

As we can see, when compared with the mathematically calculated probability, the simulation falls well within less than ±0.1% of the calculated values, confirming the probabilities calculated mathematically and empirically

**3.4 Realistic Crit Constraint from Farming**

Rather than setting an arbitrary CV threshold for "a good piece", we can find the expected best CV achievable with a fixed farming investment with an expected maximum function E[max of N independent samples]

Farming benchmark for N:

A typical two-week farming period yields:

14 days × 200 resin/day = 2,800 resin

÷ 20 resin per 5-star artifact ≈ 140 artifacts total

÷ 5 artifact types ≈ 28 pieces per slot

≈ rounded to N = 30 per artifact type.

To model the expected maximum, let X be the CV of a single randomly farmed artifact, and let x be any CV threshold value. Then, define f(x) = P(X ≤ x) as the probability that one randomly selected piece has CV at most x. f(x) is estimated from the simulation data by counting what fraction of the 5 million simulated pieces have CV ≤ x for each value of x.

If N pieces are farmed independently, the probability that none of them exceed threshold x is:

$$f(x)^N = P(MaxX \leq x)$$

This is the probability of one piece being below threshold x, multiplied by itself N times. Therefore, the probability that at least one piece exceeds threshold x is:

$$1 - f(x)^N = P(MaxX > x)$$

For any positive random variable, there is an identity that states that:

$$E[X] = \int_0^{\infty} P(X > x)\, dx$$

The idea behind this identity is that a piece with CV = 5 exceeds every threshold from x = 0 to x = 5, and no threshold beyond that. Graphing P(X > x) for this piece produces a rectangle of height 1 and width 5. Integrating this gives exactly 5, which is the CV. Essentially, integrating P(X > x) across all thresholds measures the total area over which the piece keeps exceeding threshold values, and the sum of these areas is the expected value.

Next, by substituting $P(X > x)$ , for $P(MaxX > x) = 1 - f(x)^N$ gives us

$$E[X] = \int_0^{55} (1 - f(x)^N)\, dx$$

The upper limit is 55 since that is the maximum achievable CV. This integral sums across every possible CV threshold x, the probability that the best piece after N farms exceeds it. Since each threshold contributes its width when exceeded and zero otherwise, the total area equals the expected best CV.

Since the function $f(x)$ is derived from the simulation data, and doesn't have an actual expression we can evaluate, we have to evaluate the integral in a different way. In the program, the range of the function, 0-55, is split into 2000 equal strips with a width of 0.0275. For each strip at position x, the value of $f(x)$ is looked up from the data, and the strip's value is $(0.0275)(1 - f(x)^N)$. Then, we add up the value of all the strips to find the expected value for N trials. This works because, as the width of the strips approaches zero, the sum of all the strips approaches the value of the integral.

At N = 30, the values for each slot are reported in the following table.

| Artifact, and main stat | Flower with flat HP | Feather with flat ATK | Sands with HP% | Goblet with Hydro DMG | Circlet with Crit DMG% |
|---|---|---|---|---|---|
| P(double-crit) | 32.5% | 32.5% | 32.0% | 32.0% | 31.0% |

Total expected substat CV across all five pieces:

$$32.5 + 32.5 + 32.0 + 32.0 + 31.0 = 160$$

The Circlet's CD% main stat also contributes a fixed +62.2% to CD at level 20 on a 5-star piece, but this is not a substat roll and does not count towards CV.

## 4. DAMAGE FUNCTION AND CRIT OPTIMISATION

### 4.1 Full expected damage over one Charged Attack

From gameplay observations used in this IA:

- One full channel lasts 3.0 s,
- One attack deals 8 hits of damage,
- HP decreases 5 times (once every 0.5 s),
- Each HP-loss event is 8% of max HP.

First, we can define A as the product of terms that are constant during one attack:

$$A = M \; x \; maxHP \; x \; DEFmult \; x \; RESmult$$

Let B be all non-passive DMG Bonus terms that stay constant during the attack (e.g. goblet bonus and the Marechaussee 2-piece +15% CA bonus). Let C(i) be the passive DMG Bonus at tick i. Then expected damage at tick i is:

$$E(i) = A \, x \, (1 + B + C(i)) \, x \, (1 + CR \, x \, CD)$$

So total expected damage over one Charged Attack is:

$$\sum_{i=1}^{8} A \, x \, (1 + B + C(i)) \, x \, (1 + CR \, x \, CD)$$

### 4.2 Passive talent term across the 3-second attack

The passive from Section 2, with h as the current HP fraction of max HP, is:

$$PassiveBonus(h) = 0.3 \qquad \text{for } h \geq 0.80$$

$$PassiveBonus(h) = 0.006 \times (100h - 30) \qquad \text{for } 0.30 < h < 0.80$$

$$PassiveBonus(h) = 0 \qquad \text{for } h \leq 0.30$$

Using the 5 HP-loss events of 8% each, HP during the six 0.5-second intervals is:

$$100\%, \; 92\%, \; 84\%, \; 76\%, \; 68\%, \; 60\%$$

So passive bonus values are:

$$DB = 30.0\%,\ 30.0\%,\ 30.0\%,\ 27.6\%,\ 22.8\%,\ 18.0\%$$

Since each interval has equal length (0.5 s), the attack's average passive bonus is:

$$DB_avg = (30.0 + 30.0 + 30.0 + 27.6 + 22.8 + 18.0)/6 = 26.4\%$$

Therefore, an equivalent version for expected damage is:

$$E = 8\,x\,A\,x\,(1 + B + 0.264)\,x\,(1 + totalCR\,x\,totalCD)$$

This simplification means we don't need to use 2 distinct phases when modeling the damage bonus.

### 4.3 Marechaussee Hunter bonus CR, and total CD

For this model, Marechaussee Hunter 4-piece is already fully stacked before the Charged Attack starts and is kept during the attack. So CRIT Rate gain is a constant at +36% for all 8 hits of damage. Letting c be artifact CR as a decimal, and since the character's base CR is 0.05, the total CR during the attack is:

$$CR_eff = c + 0.05 + 0.36 = c + 0.41$$

The total CD includes not only the substat CD, but two other constants as well:

$$CD_circlet_main = 0.622$$

$$CD_base_level80 = 0.788$$

This is because the circlet has a fixed CD bonus of 0.622, and Neuvillette's base CD at level 80 is 0.788. Therefore, the total CD where d is the substat CD is:

$$CD\,total = 0.622 + 0.788 + d = 1.410 + d$$

### 4.4 Crit rate vs crit damage optimisation under CV constraint

From Section 3.4, Total CV = 160 (substat CV only), and from Section 3.3, k = 2. Using decimals instead of percentages, this gives the following CV constraint:

$$2c + d = 1.60$$

where c = artifact Crit Rate and d = artifact Crit Damage from substats.

Rearranging the constraint, we find the relationship:

$$d = 1.60 - 2c$$

In the expected damage formula, the only term depending on c and d is the crit factor. After substitution for d, we see the following function for the crit multiplier:

$$F(c) = 1 + CR\ total\ x\ CD\ total$$

$$= 1 + (c + 0.41)(d + CD_fixed)$$

$$= 1 + (c + 0.41)(1.60 - 2c + 1.410)$$

$$= 1 + (c + 0.41)(3.01 - 2c)$$

$$= 2.2341 + 2.19c - 2C^2 = F(c)$$

In order to find the maximum, we can differentiate and set the derivative equal to zero:

$$F'(c) = 2.19 - 4c = 0$$

$$c = 0.5475$$

To check that this is a local maximum, we can use the Second derivative test:

$$F''(c) = -4 < 0$$

Since F" is negative, this point at c = 0.5475 is a maximum.

Next, computing the corresponding CD values using the relationship d = 1.60 - 2c:

$$d = 1.60 - 2(0.5475) = 0.505 \text{ (artifact substat CD)}$$

Plug back into total CD:

$$CD\ total = (0.505) + 1.410 = 1.915$$

This means the optimal artifact distribution is:

artifact CR = 54.75%

artifact CD (substats) = 50.5%

Total CD = 191.5%

Total CR = (0.5475) + 0.41 = 0.9575 = 95.75%

The resulting effective crit ratio is 95.75 : 191.5, which is exactly a 1 : 2 ratio. This is a well-established rule of thumb in the community when building characters; here, this ratio is derived from the crit value optimisation rather than a community convention.

**4.5 Goblet comparison: Hydro DMG% vs HP%**

Like the crit multiplier, Hydro DMG% and HP% only affect one term

$$(M\ x\ maxHP)\ \star\ (1\ +\ total\ DMG\ Bonus)$$

p = max HP% from all sources EXCEPT goblet,

B = total DMG Bonus from all sources EXCEPT goblet, where B includes the attack average passive term from 4.2.

M = talent multiplier for normal attack = 0.1251

HP = base HP

Hydro goblet multiplier ~ (0.1251(HP x (1 + p)))(1 + B + 0.466)

HP% goblet multiplier ~ (0.1251(HP x (1 + p + 0.466)))(1 + B)

To find where these provide the same bonus, set these equal:

$$(0.1251(HP\ x\ (1\ +\ p\ +\ 0.466)))(1\ +\ B)\ =\ (0.1251(HP\ x\ (1\ +\ p)))(1\ +\ B\ +\ 0.466)$$

Expanding both sides and cancelling common terms gives:

$$(1\ +\ p\ +\ 0.466)(1\ +\ B)\ =\ (1\ +\ p)(1\ +\ B\ +\ 0.466)$$

$$0.466(1\ +\ B)\ =\ 0.466(1\ +\ p)$$

$$B\ =\ p$$

Therefore, the decision is:

If B > p, HP% goblet gives higher damage.

If B < p, Hydro goblet gives higher damage.

If B = p, they are equal.

In this model, passive average is fixed at 0.264, and Marechaussee 2-piece gives another 0.15 Charged Attack DMG, so before counting other buffs:

$$B = 0.264 + 0.15 = 0.414$$

Since the sands artifact provides an initial 0.466 hp bonus:

$$p = 0.466$$

Hydro multiplier = (1 + 0.466)(1 + 0.414 + 0.466) = 1.466 x 1.880 = 2.7561

HP% multiplier = (1 + 0.466 + 0.466)(1 + 0.414) = 1.932 x 1.414 = 2.7318

Hydro Multiplier > HP% multiplier

This finding matches the break-even rule B = p. With HP% Sands, p = 0.466 exceeds B = 0.414, so adding +46.6% DMG Bonus (Hydro goblet) provides the larger advantage.

## 5. Conclusion

### 5.1 Summary of Findings

Going into this exploration, I wanted to know whether the build choices I had been making for Neuvillette were actually mathematically justified, or just a community convention that could still be optimised. The results largely confirmed what other players and guides recommend, but more importantly, they showed why those recommendations hold.

The damage model was built from mechanics that describe one full Charged Attack: 8 ticks over 3 seconds, and accounted for the HP-dependent passive bonus throughout. Rather than treating the passive as a fixed value, I tracked HP across the sequence 100%, 92%, 84%, 76%, 68%, 60%, which gave an attack average passive bonus of 26.4%.

For the goblet comparison, setting the Hydro DMG% and HP% multipliers equal produced a condition where B = p. With an HP% Sands giving p = 0.466 and a combined bonus of B = 0.414 from the passive and set bonus, the Hydro goblet came out ahead by about 0.88%, which is a small but can be a relevant advantage when every single bit of damage counts.

The crit optimisation gave what I found to be the most interesting result. Rather than assuming an arbitrary CV budget, the expected-maximum method with the constraint of N = 30 pieces per slot over two weeks, gave a total CV = 160. Optimising the crit factor under the constraint 2c + d = 1.60 gave artifact CR = 54.75% and artifact CD = 50.5%. When base stats and the Circlet main stat are included, the

full build sits at 95.75% CR and 191.5% CD, which is an exact 1:2 ratio. The fact that this emerged from the optimisation rather than being assumed as an axiom is what makes the result so interesting.

**5.2 Limitations**

The most significant limitation is how the damage model works. Enemy defence and resistance are fixed, external team buffs are excluded, and the full rotation is ignored. This was a deliberate choice to isolate the artifact optimization problems, but it means the damage model does not accurately model real gameplay. However, the crit optimization is not affected by these choices since DEF and RES affect a different term than the crit multiplier, which under normal circumstances cannot be changed.

The assumption on the Marechaussee Hunter is worth noting. The model treats the +36% CR bonus as fully stacked for all 8 ticks. In practice, there can be a short period of time at the start of an attack, roughly 0.5 seconds, or about 17% of ticks, where CR is lower than the modeled amount if Neuvillette does not successfully heal 3 times before starting an attack. This would slightly reduce the expected damage but is unlikely to meaningfully shift the optimal crit ratio.

Finally, the farming benchmark of N = 30 per slot is an estimate based on two weeks of consistent farming, and real players do not always farm artifacts like this. Total CV = 160 is a reasonable estimate for total CV rather than a ceiling.

**5.3 Extensions**

The most natural extension would be incorporating team buffs and rotation-level damage, which would allow a proper comparison of teammates like Furina, Citlali, and Sucrose rather than isolating Neuvillette alone. Beyond that, rerunning the expected-maximum model using personal artifact logs rather than simulation distributions could make the CV constraint more precise. It would also be worth testing how the results in this paper change based on different enemy resistance values and teammate buffs.

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